Coulomb completion and its classical limit: domains, collisions and preparation

For $V = -k/r$ in three dimensions, the inward radial classical flow reaches a collision in finite time, whereas the nonrelativistic Coulomb Hamiltonian $-h^2\Delta/(2m) - k/r$ on the specified Laplacian domain generates global unitary motion for every supplied reduced action $h > 0$. From rest at r_0, the classical collision time is $t_c = (\pi/2)\sqrt{mr_0^3/(2k)}$. The contrast diagnoses incomplete point-state dynamics; it does not exclude classical continuations with an added collision or state rule.

Theorem 1 separates this global operator statement from classical-limit recovery. Quadratic-force centroids obey Newton's equation exactly. For Coulomb trajectories on fixed collision-free intervals, the stated Gaussian preparations, smooth regularization and weighted localization/Hardy estimates supply the singular-potential control and position/momentum centroid bounds in §5. A norm approximation alone controls no singular multiplier. The quartic comparison likewise separates an existing fixed-action unitary limit from a classical branch rule.

The relativistic comparison in §5b has selection-dependent domains and endpoints. Dirac's finite-potential-energy choice excludes critical equality; the critical Schur form or specified cutoff limits include it. Strict-energy Klein–Gordon construction has an open endpoint, while its radial principal Friedrichs operator includes equality. A chosen radial branch is not a construction of the full Klein–Gordon evolution. These are conditional thresholds, not a necessity theorem for physical $\hbar$.

Notation: h is a supplied reduced phase constant (quantum identification $h = \hbar$, $h_P = 2\pi\hbar$), $m > 0$ the mass, $k > 0$ the force constant and $r = |x|$. Written model checks are internal; cited source theorems retain their reading labels. The planar case is a separate remark.

1. Statement

Theorem 1 (self-sufficiency contrast for Kepler's force).

(a) *Classical incompleteness.* Let $L = x \times p$ be the angular momentum. Every initial datum with $L = 0$ whose radial velocity is inward or zero, and every datum with $L = 0$ and negative energy, reaches $r = 0$ at a finite time. For a body released from rest at distance r_0 the collision time is

$$t_c = F(r_0) := \frac{\pi}{2}\sqrt{\frac{m\, r_0^3}{2k}}. \tag{1}$$

For bound data, $E < 0$ with $R = -k/E$: $t_c \le F(R)$ if the radial velocity is inward or zero, and $t_c \le 2F(R)$ if it is outward; for inward data with $E \ge 0$, $t_c \le \frac{2}{3}\sqrt{m/2k}\, r_0^{3/2}$. The collision set, on the phase space $\{(x,p) : x \neq 0\}$,

$$\mathcal{C} = \{L = 0 \text{ and } (\dot r \le 0 \text{ or } E < 0)\},$$

is nonempty, invariant under the flow while solutions exist, and of measure zero. Along every solution in $\mathcal{C}$ the force magnitude k/r^2 and the momentum $|p| \sim \sqrt{2mk/r}$ diverge as the centre is approached, so there is no continuation as a solution of the original vector field in phase space: the flow of Kepler's force is not complete. Newton's axioms, the Laws with the force law, assign no motion at the centre; any continuation is an added prescription (regularization), and the theorem establishes incompleteness, not that no classical completion is possible.

(b) *Quantum completeness.* For every $h > 0$, $m > 0$ and real k, the operator

$$H_h = -\frac{h^2}{2m}\Delta - \frac{k}{r} \qquad \text{on } L^2(\mathbb{R}^3) \tag{2}$$

is self-adjoint on the domain $H^2(\mathbb{R}^3)$ of the Laplacian and bounded below, and $U_h(t) = e^{-itH_h/h}$ is a strongly continuous unitary group defined for every $t \in \mathbb{R}$ and every initial state in $L^2(\mathbb{R}^3)$. This is Kato's theorem [@Kato1951; metadata], whose hypothesis, that the potential is the sum of an $L^2(\mathbb{R}^3)$ and a bounded function, is verified in §3.

(c) *The quartic contrast.* For $V = \lambda q^4$ on the line, $\lambda > 0$, Theorem 2(b) of the 1998-conjecture note gives, for every $h > 0$, the strong convergence of the time-sliced operators to the unitary evolution $e^{-iTH_h/h}$ along every sequence of partitions, with no extra datum; Theorem 1 there shows that the unrestricted halved classical-first construction at zero resolution has no limit on a two-cell partition with three stationary paths, and Theorem 2(a) there exhibits one repair, a branch selection chosen by hand. The contrast is between the fixed-h construction, which exists for all data, and that specific classical-first prescription, which needs a datum the action does not supply; it says nothing against the quartic Newtonian initial-value problem itself, which is globally well posed.

(d) *The classical trajectory as a derived object.* (d1) For $H = p^2/2m + V$ with $\deg V \le 2$, the expectations $\langle q\rangle_t$, $\langle p\rangle_t$ of every state with finite second moments obey Newton's equations exactly for every $h > 0$,

$$\frac{d}{dt}\langle q\rangle_t = \frac{\langle p\rangle_t}{m}, \qquad \frac{d}{dt}\langle p\rangle_t = -V'(\langle q\rangle_t), \tag{3}$$

and a coherent state stays a Gaussian whose centroid is the classical trajectory. (d2) For Kepler's force, let $z_0 = (x_0, p_0)$ lie on a classical orbit whose distance from the centre stays at least $\rho_{\min} > 0$ on $[0, T]$. Fix $0 < \delta < \rho_{\min}$ and a real $V_\delta \in C_b^\infty$ equal to $-k/r$ for $r \ge \delta$. For the normalized coherent preparations of §5a, with width $\ell_h = \sqrt{hL_*/P_*}$ and fixed $L_*, P_* > 0$, the regularized and Coulomb evolutions both satisfy the Gaussian norm approximation (4) with error $O(\sqrt{h})$, uniformly on $[0, T]$. Section 5a proves

$$\sup_{[0,T]} \|(V - V_\delta)U_h^{(\delta)}(t)\varphi_h\|_2 = O(h^{3/2}) = o(h), \qquad \text{(L)}$$

and separately proves that the Coulomb position and momentum centroids are $q_t + O(\sqrt{h})$ and $p_t + O(\sqrt{h})$. Here "large action" means this fixed-scale limit $h \to 0$ at fixed orbit. All constants depend on the fixed interval, regularization and preparation; no uniformity in T or δ is claimed.

Conclusion. The theory with $h > 0$ defines motion for all data and all times, by (b), and yields Newton's trajectories where they exist, exactly for quadratic forces and, under the fixed-data hypotheses of §5a, for Kepler's force away from collisions, by (d), while Newton's axioms leave motion undefined on the collision set, by (a), and the unrestricted classical-first construction at zero needs a datum the action does not supply, by (c).

2. Proof of (a)

Conservation of L under a central force is Proposition I of Book I; for $L = 0$ the motion is on a line through the centre and $E = \frac{m}{2}\dot r^2 - k/r$ is conserved. Released from rest at r_0, $E = -k/r_0$ and

$$\dot r^2 = \frac{2k}{m}\Big(\frac{1}{r} - \frac{1}{r_0}\Big) = \frac{2k}{m}\,\frac{r_0 - r}{r\,r_0},$$

with $\dot r < 0$ for $t > 0$ because the force points inward. Hence

$$t_c = \int_0^{r_0} \frac{dr}{|\dot r|} = \sqrt{\frac{mr_0}{2k}} \int_0^{r_0} \sqrt{\frac{r}{r_0 - r}}\, dr.$$

With $r = r_0 \sin^2\theta$, $dr = 2r_0 \sin\theta\cos\theta\, d\theta$ and $\sqrt{r/(r_0 - r)} = \tan\theta$, the integral is $2r_0 \int_0^{\pi/2} \sin^2\theta\, d\theta = \pi r_0/2$, which gives (1).

For a bound datum with $L = 0$, $E < 0$ and $R = -k/E$: $r \le R$ throughout, $|\dot r|^2 = \frac{2k}{m}(1/r - 1/R)$, and the time from r_0 to the centre along the inward branch is $I(r_0) = \sqrt{m/2k}\int_0^{r_0} \sqrt{rR/(R-r)}\, dr \le I(R) = F(R)$. If the radial velocity is outward, the body first rises to R in the time $F(R) - I(r_0)$ and then falls in the time $F(R)$, so its collision time is $2F(R) - I(r_0) \le 2F(R)$, and it exceeds $F(R)$: one fall time does not bound it. For inward data with $E \ge 0$, $|\dot r| \ge \sqrt{2k/(mr)}$ gives the time bound $\int_0^{r_0} \sqrt{mr/2k}\, dr = \frac{2}{3}\sqrt{m/2k}\, r_0^{3/2}$. Invariance of $\mathcal{C}$: L and E are conserved; for $E < 0$ the condition is then automatic, and for $E \ge 0$ the radial velocity cannot vanish, since $\frac{m}{2}\dot r^2 = E + k/r > 0$, so its inward sign is preserved. The condition $L = 0$ is that p be parallel to x, which for $x \ne 0$ defines a four-dimensional subset of the six-dimensional phase space, hence of Liouville measure zero; it is nonempty. As $r \to 0$ along a solution in $\mathcal{C}$, the force magnitude k/r^2 is unbounded and $\frac{m}{2}\dot r^2 = E + k/r$ gives $|p| \sim \sqrt{2mk/r}$, so the solution leaves every compact subset of phase space in finite time and admits no continuation as a solution of the vector field. A continuation through the centre (Levi-Civita or Kustaanheimo–Stiefel regularization) is an added prescription with an enlarged notion of solution, in the same position as the branch rule of (c). □

3. Proof of (b), with Kato's hypothesis verified

Kato's theorem [@Kato1951; metadata; the statement used is the one standardly quoted from it, the primary text was not read in this session]: let $V = V_1 + V_2$ be real with $V_1 \in L^2(\mathbb{R}^3)$ and $V_2 \in L^\infty(\mathbb{R}^3)$. Then for every $a > 0$ the operator $-a\Delta + V$ is self-adjoint on $H^2(\mathbb{R}^3)$ and bounded below. The mechanism is the Sobolev bound $\|u\|_\infty \le \epsilon\|\Delta u\| + C_\epsilon\|u\|$ for $u \in H^2(\mathbb{R}^3)$, which makes V_1 relatively bounded with respect to Δ with relative bound zero, so that the Kato–Rellich theorem applies for every value of $a = h^2/2m$.

Verification for $V = -k/r$: fix a cutoff radius $R > 0$ and put $V_1 = -\frac{k}{r}\mathbf{1}_{r<R}$, $V_2 = -\frac{k}{r}\mathbf{1}_{r\ge R}$. Then $|V_2| \le |k|/R$, and

$$\int_{\mathbb{R}^3} |V_1|^2 d^3x = k^2 \int_0^R \frac{4\pi r^2}{r^2}\, dr = 4\pi k^2 R < \infty.$$

So the hypothesis holds for every real k, and H_h in (2) is self-adjoint on $H^2(\mathbb{R}^3)$ for every $h > 0$, $m > 0$. Stone's theorem then gives the unitary group $U_h(t)$ for all $t \in \mathbb{R}$, strongly continuous, defined on every $\psi \in L^2$, with $U_h(t)\psi \in H^2$ for $\psi \in H^2$; states outside the operator domain have the mild unitary evolution, those inside satisfy the Schrödinger equation. No initial datum and no time is excluded; the states supported near $r = 0$ evolve like all others. □

Planar remark. In $\mathbb{R}^2$ the function $1/r$ is not square integrable near the origin, and the operator-sense theorem does not apply; the planar Coulomb Hamiltonian is defined in the form sense by the KLMN theorem, since $1/r \in L^p_{\mathrm{loc}}(\mathbb{R}^2)$ for $p < 2$ is form-bounded relative to $-\Delta$ with relative bound zero. The planar Kepler problem used elsewhere in this repository therefore also has a global unitary dynamics, by that standard route, which is not reproved here.

4. Proof of (c)

The statements are Theorems 1, 2(a) and 2(b) of the 1998-conjecture note, accepted on refereeing there (status line of that note). Theorem 2(b) proves the product formula for $H_h = -\frac{h^2}{2m}d^2/dq^2 + \lambda q^4$ at every $h > 0$: essential self-adjointness, the strong limit of the symmetric time slicing along any partition sequence, and the composition $\|U_h(T)\| = 1$. Theorem 1 proves that the halved classical-first functional on the two-cell partition has, as the resolution tends to zero, the oscillating value (4) there with no limit, because the three stationary paths interfere; Theorem 2(a) restores a limit only by restricting to a neighbourhood of one path, a datum the action

does not contain. Hence for the quartic the $h > 0$ theory exists for all data without extra rules and the construction at zero does not. □

5. Proof of (d)

(d1), quadratic forces. By Theorem A(a) of the fifth-postulate note, the Heisenberg equations are Newton's second law as operator identities for every real h: $\dot q = p/m$ and $\dot p = -V'(q)$. Taking expectations in a state with finite second moments gives $\frac{d}{dt}\langle q\rangle = \langle p\rangle/m$ and $\frac{d}{dt}\langle p\rangle = -\langle V'(q)\rangle$; for $\deg V \le 2$ the function V' is affine, so $\langle V'(q)\rangle = V'(\langle q\rangle)$, which is (3). The centroid therefore follows the classical trajectory through $(\langle q\rangle_0, \langle p\rangle_0)$ for every $h > 0$ and every such state, with no approximation. For a coherent state the Wigner function is the Gaussian transported by the affine flow (proof of Theorem 5 of the reachability note), so the state stays Gaussian with the classical trajectory as its centroid and covariance $M_t \Sigma_h M_t^{\mathsf T}$. The width at time t depends on the flow: for free motion, scaling the mass by λ at fixed h and widths sends $M_{12} = t/m \to 0$ and the position width to the initial σ, while for Hooke's force at fixed frequency the width $s_t^2 = \sigma^2 \cos^2 \omega t + h^2 \sin^2 \omega t/(4m^2\omega^2\sigma^2)$ tends to $\sigma^2 \cos^2 \omega t$; the centroid transport is exact in every case. □

5a. Kepler localization and moments (2026-10-02)

Target and assumptions. Fix $m, k, L_*, P_* > 0$, an initial phase point $z_0 = (q_0, p_0)$, and $T < \infty$ such that its Kepler orbit $z_t = (q_t, p_t)$ exists on $[0, T]$ and $\rho_{\min} := \min_{[0,T]} |q_t| > 0$. Fix $0 < \delta < \rho_{\min}$ and a real $V_\delta \in C_b^\infty(\mathbb{R}^3)$ equal to $-k/r$ on $r \ge \delta$. All these choices are independent of h. Put $\ell_h = \sqrt{hL_*/P_*}$, choose $h_0 > 0$, and consider $0 < h \le h_0$ with the normalized preparations

$$\varphi_h(x) = (2\pi\ell_h^2)^{-3/4} \exp\left[-\frac{|x - q_0|^2}{4\ell_h^2} + \frac{ip_0 \cdot (x - q_0)}{h}\right].$$

Both Hamiltonians have domain $H^2(\mathbb{R}^3)$. The conclusions to prove are (L), its Duhamel transfer, and convergence of the two first moments. Neither singular-multiplier control from an L^2 approximation nor convergence of an unbounded observable is an assumption.

Primary Hepp statement. The published paper [@Hepp1974; passage, §2, Theorem 2.1, p. 268; equations (2.15)–(2.20), pp. 269–270; Theorem 2.3 and condition (R), p. 271] states Theorem 2.1 for one degree of freedom: a real potential, $C^{2+\epsilon}$ near the classical orbit, $\epsilon > 0$, with $\int |V(x)|^2 e^{-\eta x^2} dx < \infty$ for some $0 < \eta < \infty$, and any self-adjoint realization of the scaled Hamiltonian. The limits are strong limits of Weyl operators, uniformly on compact time intervals inside the orbit's existence interval. Equation (2.20) gives the corresponding norm approximation on Gaussian packets. For finitely many coordinates, Theorem 2.3 replaces the integrability assumption by (R): a strongly continuous unitary propagator $U_h(t, s)$ for all real t, s, with $U_h(t, s)U_h(s, r) = U_h(t, r)$, $U_h(t, s)^* = U_h(s, t)$, $U_h(t, t) = I$, and the strong derivative

$$\partial_s U_h(t, s)\psi = \frac{i}{h} U_h(t, s) H_h(s)\psi$$

on the sufficiently decaying Gaussian packets specified in (2.27), with $H_h(s)\psi$ defined as a differential operator. Theorem 2.3 allows finitely many particles with vector and scalar potentials locally $C^{2+\epsilon}$ along the orbit; here the vector potential is zero.

The three-dimensional V_δ satisfies (R) because it is bounded and its Gaussian packets belong to H^2. The approximation is

$$\sup_{0 \le t \le T} \|U_h^{(\delta)}(t)\varphi_h - e^{iS_t/h}\varphi^h_{z_t, M_t}\|_2 = O(\sqrt{h}), \tag{4}$$

where $S_t = \int_0^t (p_s \cdot \dot{q}_s - H_{\mathrm{cl}}(z_s))ds$ and the Gaussian is obtained with the continuously lifted quadratic propagator of the linearized flow M_t. Its translation convention uses $p_t \cdot (x - q_t)$ in the phase. To see the rate in this smooth case, expand V_δ through second order at q_t: the cubic remainder has norm at most $B_3 \ell_h^3 \| |y|^3 g_t \|_2 / 6$ on the centered quadratic Gaussian g_t. Duhamel divides this by h and integrates over $[0, T]$; the Gaussian moment is bounded on that fixed interval. This is a direct specialization of Hepp's proof, not an estimate of the singular multiplier on the unknown norm remainder.

Centered Sobolev bound. Let $u_h(t) = U_h^{(\delta)}(t)\varphi_h$ and write exactly

$$u_h(t,x) = \ell_h^{-3/2} e^{i[S_t + p_t \cdot (x - q_t)]/h} a_h(t,y), \qquad y = (x - q_t)/\ell_h.$$

Hamilton's equations cancel the constant and linear terms, so

$$i\partial_t a_h = G_h(t) a_h, \quad G_h = -a_* \Delta_y + W_h(t,y), \quad a_* = \frac{P_*}{2mL_*},$$

$$W_h = \frac{V_\delta(q_t + \ell_h y) - V_\delta(q_t) - \ell_h \nabla V_\delta(q_t) \cdot y}{h}.$$

Set $B_j = \sup_x \|\nabla^j V_\delta(x)\|$ (the Euclidean tensor norm), $\ell_0 = \sqrt{h_0 L_* / P_*}$, and, for integer $s \geq 1$,

$$b_s = \frac{L_*}{P_*} \max_{2 \leq j \leq \max(2,s)} \ell_0^{j-2} B_j.$$

Taylor's integral formula gives $|\nabla_y W_h| \leq b_s |y|$, and $|\partial_y^\gamma W_h| \leq b_s$ for $2 \leq |\gamma| \leq s$. All these constants are uniform for $0 < h \leq h_0$ and $0 \leq t \leq T$.

With $D_y = -i\nabla_y$, define

$$E_s(a) = \sum_{|\alpha| + |\beta| \leq s} \| y^\alpha D_y^\beta a \|_2^2, \quad N_s = \binom{s+6}{6}, \quad c_s = a_* s(s+1) + b_s(2^s - 1 + 2s).$$

The commutator of $y^\alpha D_y^\beta$ with the kinetic term has norm bounded by $a_* s(s+1)\sqrt{E_s}$. For the potential term use

$$[D_y^\beta, W_h] = \sum_{0 < \gamma \leq \beta} \binom{\beta}{\gamma} (-i)^{|\gamma|} (\partial^\gamma W_h) D_y^{\beta - \gamma}.$$

When $|\gamma| = 1$, the bound $b_s |y|$ increases the position degree by one and decreases the derivative degree by one. Bounding $|y|$ by $\sum_j |y_j|$ gives $3 s b_s \sqrt{E_s}$. The remaining binomial coefficients sum to at most $2^s - 1 - s$. No term has degree larger than s. Thus

$$\frac{d}{dt} E_s(a_h) \leq 2 c_s \sqrt{N_s}\, E_s(a_h), \qquad E_s(a_h(t)) \leq E_s(a_0) e^{2 c_s \sqrt{N_s} T}, \tag{4a}$$

where $a_0 = (2\pi)^{-3/4} e^{-|y|^2/4}$. For fixed h the calculation is justified on Schwartz packets: the integral equation for a bounded smooth potential, with position and derivative cutoffs, gives the same finite-seminorm energy estimates and propagation of Schwartz regularity; letting the cutoffs tend to infinity proves (4a). Equivalently, one first obtains these estimates for smooth truncated data and passes by the L^2 evolution and lower semicontinuity. The centered estimates, unlike the preliminary fixed-h ones, have the uniform constants displayed above.

For use below define

$$A_s(T) = 3^{s/2} \sqrt{E_s(a_0)}\, e^{c_s \sqrt{N_s} T}.$$

Expanding $|y|^{2n} = (y_1^2 + y_2^2 + y_3^2)^n$ shows that both $\| |y|^n a_h(t) \|_2$ and $\| |y|^n \nabla_y a_h(t) \|_2$ are at most $A_{n+1}(T)$.

Proof of (L). Put $r_c = (\delta + \rho_{\min})/2$ and $d = \rho_{\min} - r_c > 0$. Choose a smooth radial cutoff χ equal to one for $r \le \delta$, zero for $r \ge r_c$, with $0 \le \chi \le 1$ and $\|\nabla\chi\|_\infty \le 2/(r_c - \delta) = 2/d$. On its support $|x - q_t| \ge d$, hence $|y| \ge d/\ell_h$. The centered bounds imply, for every integer $n \ge 1$,

$$\|\chi u_h(t)\|_2 \le A_{n+1}(T)(\ell_h/d)^n, \qquad \|\chi(\nabla - ip_t/h)u_h(t)\|_2 \le A_{n+1}(T)\ell_h^{n-1}/d^n.$$

Apply the three-dimensional Hardy inequality $\|f/r\|_2 \le 2\|\nabla f\|_2$ to $f = e^{-ip_t\cdot x/h}\chi u_h(t)$; the phase removes the potentially large carrier momentum from the estimate. Since $V - V_\delta$ vanishes outside $r < \delta$, this gives

$$\sup_{0\le t\le T} \|(V - V_\delta)u_h(t)\|_2 \le \frac{A_{n+1}(T)}{d^n}\left[(4k/d + B_0)\ell_h^n + 2k\ell_h^{n-1}\right]. \tag{4b}$$

For $n = 4$ the right side is $O(h^{3/2}) = o(h)$, proving (L). Choosing larger n gives any prescribed power of h. The dependence is explicit through T, m, L_*/P_*, the preparation seminorms, $B_j(V_\delta)$, and the separation $d = (\rho_{\min} - \delta)/2$. No uniformity as $T \to \infty$, $\delta \to 0$, or $\rho_{\min} - \delta \to 0$ is claimed. A bounded family of centered Schwartz preparations satisfies the same estimate with its uniform $E_s(a_0)$ in place of the Gaussian value. For the usual scaled regularization $V_\delta(x) = -(k/\delta)v(x/\delta)$, with fixed smooth bounded v equal to $1/|x|$ for $|x| \ge 1$, the derivative dependence is $B_j \le k\delta^{-j-1}\|\nabla^j v\|_\infty$. A radius δ alone cannot bound these constants for arbitrary regularizations.

On the common domain H^2, Duhamel now gives

$$U_h(t)\varphi_h - U_h^{(\delta)}(t)\varphi_h = -\frac{i}{h}\int_0^t U_h(t-s)(V - V_\delta)u_h(s)\,ds.$$

Consequently the supremum of this norm is $O(h^{(n-3)/2})$ for every $n \ge 4$. In particular the two evolutions differ by $O(h^M)$ for every fixed $M > 0$, with an M-dependent constant. Equation (4) therefore transfers to the Coulomb evolution.

First moments. Write $\psi_h = U_h\varphi_h$ and $p = -ih\nabla$. Its conserved form energy e_h is bounded independently of h: the initial kinetic energy is $|p_0|^2/(2m) + 3hP_*/(8mL_*)$, and the elementary spherical Gaussian integral gives $\langle\varphi_h, r^{-1}\varphi_h\rangle = \operatorname{erf}(|q_0|/(\sqrt{2}\ell_h))/|q_0| \le 1/|q_0|$. Let $e_* = \sup_{0<h\le h_0}|e_h| < \infty$. Hardy and Young's inequality give, with $X_h(t) = \|p\psi_h(t)\|_2$,

$$e_h \ge \frac{X_h(t)^2}{2m} - \frac{2k}{h}X_h(t) \ge \frac{X_h(t)^2}{4m} - \frac{4mk^2}{h^2}, \qquad X_h(t) \le 2\sqrt{me_*} + \frac{4mk}{h}. \tag{4c}$$

The continuity equation, first with bounded position cutoffs and then by passage to the limit, gives

$$\|x\psi_h(t)\|_2 \le \|x\varphi_h\|_2 + \frac{1}{m}\int_0^t X_h(s)\,ds = O(1 + h^{-1}). \tag{4d}$$

This also proves preservation of the position-moment domain on the fixed interval. Energy conservation preserves the Coulomb form domain H^1, so the momentum norm is finite as well. For u_h the centered bounds give $\|(x - q_t)u_h(t)\|_2 \le \ell_h A_1(T)$ and $\|(p - p_t)u_h(t)\|_2 \le (h/\ell_h)A_1(T)$.

For each self-adjoint coordinate or momentum component B and the two normalized states, symmetry on its domain yields

$$|\langle\psi_h, B\psi_h\rangle - \langle u_h, Bu_h\rangle| \le \|\psi_h - u_h\|_2(\|B\psi_h\|_2 + \|Bu_h\|_2).$$

Use (4b) with $n = 6$: the norm difference is $O(h^{3/2})$, while (4c)–(4d) bound the second factor by $O(h^{-1})$. Thus, uniformly on $[0, T]$,

$$\langle x\rangle_{\psi_h(t)} = q_t + O(\sqrt{h}), \qquad \langle p\rangle_{\psi_h(t)} = p_t + O(\sqrt{h}). \tag{4e}$$

These first-moment statements use derivatives of V_δ through order seven via (4a). They establish the centroid assertion without assuming moment convergence from a norm approximation. They give neither long-time control nor convergence of higher moments. □

5b. Proposition 2: action thresholds under specified selections

Theorem 1(b) gives no positive lower threshold: the non-relativistic Coulomb evolution exists for every $h > 0$. At finite c a threshold depends on which relativistic model, domain and selection premise are required. Fix $m, k, c > 0$ and $\nu = k/(hc) > 0$. All statements concern an ideal point Coulomb source. Rescale length by $\lambda_h = h/(mc)$ and energy by mc^2; in the formulas for domains below, x and r denote these dimensionless coordinates. This unitary rescaling leaves ν unchanged.

Primary sources and their scope. Schmincke's original example, p. 348, applies his Theorems 2–3 to the strict range $|\nu| < 1$ and identifies the selected domain by finite potential energy [@Schmincke1972; passage, pp. 345–348, GDZ text]. Wüst's theorem and corollary require a real C^1 potential dominated by μ/r with $\mu < 1$, together with the specified bounded cutoff and convergence conditions; the selected domain is $D(D^*_{\min}) \cap D(r^{-1/2})$ [@Wuest1975; passage, pp. 93–95, GDZ text]. Those results do not settle $\nu = 1$. Nenciu constructs the finite-kinetic-form-domain extension for a strict subcritical Coulomb bound [@Nenciu1976; passage, introduction and Theorem 5.1, pp. 246–247, author-posted text]; its poorly extracted formulas are not used here to infer an endpoint.

The precise subcritical domain statement used below is Theorem 1.1 of Esteban–Lewin–Séré: for real $V = V_1 + V_2 + V_3$ with $|V_1| \le \nu/r$, $0 \le \nu < 1$, $V_2 \in L^3$ and $V_3 \in L^\infty$, there is a unique extension with $H^1 \subset D(D_V) \subset H^{1/2}$, also uniquely characterized by $\int |\Psi|^2/r < \infty$ for every domain vector [@EstebanLewinSere2019; passage, arXiv v7, Theorem 1.1, pp. 9–10]. Their §1.2 and Appendix A include essential self-adjointness at $\nu = \sqrt{3}/2$. Their Theorem 1.11, building on Esteban–Loss [@EstebanLoss2007; passage, Definition 3 and Theorem 4, pp. 3–4], supplies a different critical domain and norm-resolvent cutoff limit.

For Klein–Gordon, Case's bibliographic identity is verified as *Physical Review* **80**, 797–806 (1950), DOI 10.1103/PhysRev.80.797 [@Case1950; metadata/abstract]. Its original body was not accessible in this revision, so it is not a proof input. We derive the radial indices below directly and use separate read sources for the domains and the evolution. Domain and spectral facts for the inverse-square principal operator come from Dereziński–Richard [@DerezinskiRichard2017; passage, §§2.2–2.3, equations (2.3)–(2.6), Corollary 2.4 and Theorem 5.5]. The actual global Klein–Gordon construction used here is Langer–Najman–Tretter [@LangerNajmanTretter2008; passage, Example 8.3 and Theorems 8.5–8.6, pp. 744–747]. These are distinct results.

Proposition 2 (selection-dependent admissible sets). The following sets, including their endpoints, correspond to the premises in the first column. An admissible set in one row is not an exclusion theorem for other rows or other completions.

Selection premise	Admissible h-set	Result supplied
Dirac minimal operator on $C_c^\infty(\mathbb{R}^3 \setminus \{0\})^4$: require essential self-adjointness	$[2k/(\sqrt{3}c), \infty)$	Unique closure, global unitary evolution
Dirac: require every operator-domain vector to have finite separate Coulomb potential energy	$(k/c, \infty)$	Unique distinguished extension, global unitary evolution
Dirac: use that extension below critical coupling and the critical Schur-domain or specified cutoff-limit prescription at equality	$[k/c, \infty)$	Global unitary evolution, including $h = k/c$

Selection premise	Admissible h-set	Result supplied
Dirac: permit additional self-adjoint boundary conditions in singular channels	$(0,\infty)$	Extensions exist; uniqueness is lost where the minimal operator is not essentially self-adjoint
Klein–Gordon: require strict coercivity of the conserved energy in the usual $H^1 \oplus L^2$ topology, and use the charge-space generator below	$(2k/c,\infty)$	A global charge-space evolution is constructed; the strict criterion excludes equality
Klein–Gordon s-wave principal radial operator: choose its Friedrichs extension	$[2k/c,\infty)$	Principal radial operator and branch, including the no-log branch at equality; full critical evolution is not concluded

Dirac proof and domains. In the scaled coordinates the expression is $D_\nu = -i\alpha\cdot\nabla+\beta-\nu/r$. In a channel $\kappa \in \mathbb{Z}\setminus\{0\}$ its reduced two-component radial operator acts in $L^2((0,\infty),dr)^2$ as

$$d_{\nu,\kappa} = \begin{pmatrix} 1-\nu/r & -\partial_r+\kappa/r \\ \partial_r+\kappa/r & -1-\nu/r \end{pmatrix}.$$

Its origin modes have exponents $\pm s_\kappa$, $s_\kappa = \sqrt{\kappa^2-\nu^2}$. For real $s_\kappa > 0$, the less regular mode is locally square integrable precisely when $s_\kappa < 1/2$; for imaginary s_κ both modes are locally square integrable. At $s_\kappa = 1/2$ its squared norm contains $\int dr/r$ and diverges. Infinity is limit point. Taking $|\kappa| = 1$ gives the inclusive essential self-adjointness threshold $\nu \le \sqrt{3}/2$. The closure has domain H^1 for strict inequality; at equality its domain is larger than H^1 (Appendix A, Proposition A.1 of the cited source).

For $\sqrt{3}/2 < \nu < 1$, both modes are in radial L^2. A full spinor contains the additional factor $1/r$, so its potential integral is $\int_0^\epsilon (|u|^2+|v|^2)dr/r$. Finite potential energy excludes the r^{-s_1} coefficient and retains r^{s_1}, giving the unique domain $D(D^*_{\min})\cap D(r^{-1/2})$. At $\nu = 1$ the modes are constant and logarithmic. Both have divergent separate potential energy. An extension domain must contain a nonzero boundary mode: setting both coefficients to zero leaves the minimal, non-self-adjoint domain. Thus no self-adjoint extension at equality has all domain vectors of finite separate potential energy. For $\nu > 1$ the lowest channel has imaginary s_1 and the same integral diverges for every nonzero boundary mode. This proves the open set in the second row.

The critical prescription instead uses the closed Schur form of Esteban–Loss. For $0 < \gamma < 1$, define on $C_c^\infty(\mathbb{R}^3,\mathbb{C}^2)$

$$b_\gamma[\phi] = \int_{\mathbb{R}^3} \left[\frac{r}{1+\gamma r}|\sigma\cdot\nabla\phi|^2 + (2-\gamma-r^{-1})|\phi|^2 \right] dx.$$

Their critical Hardy–Dirac inequality (7) verifies Assumption (A) with $c(V) = 0$ and $\Gamma = \sup V = 0$. Let $\mathcal{H}_+$ be the form domain of its closure, equivalently the completion in $(b_\gamma[\phi] + \|\phi\|_2^2)^{1/2}$. Proposition 2 makes this domain independent of γ in the stated range. The closure defines the combined form; its two integrals need not exist separately for a completed vector. Definition 3 and Theorem 4 give the operator domain

$$\mathcal{D}_C = \{\Psi = (\phi,\chi) \in L^2(\mathbb{R}^3)^4 : \phi \in \mathcal{H}_+,\ D_1\Psi \in L^2 \text{ in distributions}\}.$$

They prove self-adjointness and uniqueness under that form-domain premise. The angular-channel formulation in Esteban–Lewin–Séré, equations (1.28)–(1.31), admits $1/r$ spinor singularities and excludes $(\log r)/r$ in both lowest channels. Its combined operator energy is finite;

its separate potential and kinetic form energies need not be. Their Theorem 1.11, for $V \geq -1/r$ and $\sup V < 1$, reports norm-resolvent convergence for $V_\epsilon = \max(V, -1/\epsilon)$ or $V_\epsilon = (1-\epsilon)V$, as $\epsilon \downarrow 0$. This establishes the included endpoint under this changed selection premise.

Source-formula clarification, 2026-10-02. The literal compact weighted-gradient formula printed in (1.32) of arXiv v7 is not used as a domain definition here: it conflicts with the angular form description just cited. For a nonzero constant spinor ξ, put

$$\phi(x) = \frac{e^{-r}}{r}\sigma \cdot \omega_x \xi, \qquad \omega_x = x/r.$$

Its critical radial profile $\phi_1 = e^{-r}\xi/r$ satisfies $r\phi_1' + \phi_1 = -e^{-r}\xi$, so the angular form norm in (1.29) is finite. Logarithmic cutoffs of $r\phi_1$ have cost $\int r|\eta_\epsilon'|^2 dr = 1/\log(1/\epsilon)$ on $[\epsilon^2, \epsilon]$ and place it in the Schur-form closure. But

$$\sigma \cdot \nabla(r\phi) = e^{-r}(2/r - 1)\xi,$$

whose squared norm with weight $1/[r(1+r)]$ diverges like $\int_0 dr/r$. Thus that compact rewrite cannot be taken literally as equivalent to the angular norm. The source TeX and HTML confirm that this is not a PDF extraction change. We use the original Esteban–Loss form closure and the explicit angular conditions; this calculation checks the problematic rewrite, not the full borrowed theorems independently.

For arbitrary $\nu > 0$, conjugation of the real radial system makes its two deficiency indices equal. Every limit-circle channel therefore admits self-adjoint boundary conditions; only finitely many channels require them for a fixed ν. Their direct sum gives a global self-adjoint Dirac realization. Above critical coupling an origin phase is additional data. Self-adjointness alone consequently gives no positive lower bound on h.

Klein–Gordon construction and domains. In the scaled variables the equation is $[(i\partial_t - V)^2 + \Delta - 1]\psi = 0$, $V = -\nu/r$. Hardy's inequality gives $\|Vf\|_2 \leq 2\nu\|\nabla f\|_2$ for $f \in H^1$. Langer–Najman–Tretter's Theorem 8.5 assumes $H^1 \subset D(V)$ and a decomposition $V = V_0 + V_1$ with

$$\|V_0 f\|_2 \leq a\|f\|_2 + b\|(-\Delta + 1)^{1/2} f\|_2, \quad a + b < 1,$$

and V_1 relatively compact. Pure Coulomb satisfies it with $a = 0$, $b = 2\nu < 1$, $V_1 = 0$. The theorem constructs the generator

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} Vx + y \\ (-\Delta + 1)x + Vy \end{pmatrix}$$

in the charge space $\mathcal{K} = H^{1/2} \oplus H^{-1/2}$, with domain

$$D(A) = \{(x, y) \in \mathcal{K} : x \in H^1,\ y \in L^2,\ Vx + y \in H^{1/2},\ (-\Delta + 1)x + Vy \in H^{-1/2}\}.$$

The charge pairing is $[X, X'] = \langle x, y' \rangle + \langle y, x' \rangle$, using Sobolev duality. The group is strongly continuous and unitary for this indefinite pairing. It gives global mild evolution on $\mathcal{K}$ and classical evolution on $D(A)$. It is not a probability-unitary group on scalar L^2. Theorem 8.6 gives the corresponding strong Klein–Gordon solutions for $\psi_0 \in H^1$, $\psi_1 \in H^{1/2}$ and $(-\Delta - V^2)\psi_0 \in H^{-1/2}$. For the sign convention of the equation above use $i\partial_t X = AX$, $y = (i\partial_t - V)\psi$, and e^{-itA}; the source's group e^{itA} also gives this evolution after reversing the time parameter.

In physical units the conserved energy has spatial part

$$Q_\nu[f] = h^2c^2\|\nabla f\|_2^2 + m^2c^4\|f\|_2^2 - k^2\|f/r\|_2^2 \geq (1 - 4\nu^2)h^2c^2\|\nabla f\|_2^2 + m^2c^4\|f\|_2^2.$$

Sharpness of Hardy's constant, with dilations of its near-extremizers to remove the mass term relative to the gradient, shows that strict coercivity in the usual H^1 norm holds exactly for $\nu < 1/2$. At equality the combined form is nonnegative but loses that coercivity; above it the

spatial form is unbounded below. This settles the fifth row's criterion, not the existence of all possible Klein–Gordon completions.

For a stationary s-wave $\psi = e^{-iEt/h}u(r)/r$, direct substitution gives the pencil

$$-h^2c^2u'' + (m^2c^4 - E^2)u - \frac{2Ek}{r}u - \frac{k^2}{r^2}u = 0.$$

The indices are $1/2 \pm s$, $s = \sqrt{1/4 - \nu^2}$. The principal operator $B_\nu = -\partial_r^2 - \nu^2/r^2$ starts on $C_c^\infty(0,\infty)$ in $L^2(dr)$, with maximal domain consisting of u, u' locally absolutely continuous and $B_\nu u \in L^2$. For $0 < \nu < 1/2$ its Friedrichs domain sets the coefficient of $r^{1/2-s}$ to zero. Both branches have finite $u(0)$ and are locally L^2; those properties alone do not select it. At $\nu = 1/2$ the modes are $\sqrt{r}$ and $\sqrt{r}\log r$. The Friedrichs domain sets the logarithmic coefficient to zero, admitting the endpoint for this principal operator. It permits a nonzero $\sqrt{r}$ coefficient whose separate kinetic integral diverges; the critical combined form is finite: writing $u = \sqrt{r}\,v$ on $C_c^\infty(0,\infty)$ gives $\int(|u'|^2 - |u|^2/(4r^2))dr = \int r|v'|^2 dr$. A logarithmic cutoff near zero approximates a constant v with vanishing form cost. For $\nu > 1/2$, $s = i\beta$, self-adjoint extensions require an origin phase and all have a negative spectrum unbounded below (Dereziński–Richard, Theorem 5.5(ii)). No semibounded Friedrichs extension exists there. The lower-order $1/r$ term of the stationary pencil does not change the indicial classification.

Uncontrolled critical dynamical term. The no-log radial selection does not prove a global Klein–Gordon group at $\nu = 1/2$. The critical combined form does not bound $\|f/r\|_2$: a cutoff of $\sqrt{r}$ in the radial form domain already has infinite separate kinetic and inverse-square integrals. Thus the multiplier bound with relative constant strictly below one fails. On the enlarged critical form space, the mixed dynamical term $V\partial_t\psi$, or equivalently the required control of $VQ_{1/2}^{-1/2}$, remains unbounded by the estimates used here. A compatible phase space and generator domain are still needed for a critical full evolution theorem. No such theorem is asserted by the sixth row. □

Physical selection premises. Dirac self-adjointness ensures probability conservation, independently of the action-threshold argument. It does not select among its extensions. Requiring each separate energy to be finite is an additional regularity premise; finite total energy does not entail it, as the critical domain shows. The specified bounded-source and subcritical-coupling limits give independent mathematical support for the critical Dirac prescription once those preparations are chosen. They do not prove that all source regularizations have that limit or that the ideal point source is mandatory. Klein–Gordon energy stability supports semibounded principal forms; Friedrichs closure and a particular topology are additional choices, and charge conservation is not positive probability conservation. A finite-size source introduces further data and is outside these point-source thresholds.

The unit k/c agrees with the necessity-unit note and the relativistic Kepler threshold. That dimensional agreement supplies neither a selection principle nor the measured value of h. In particular there is no single unqualified set $[h_{\min}, \infty)$ shared by these models.

6. What the theorem establishes and what it does not

Theorem 1 compares an incomplete classical point-force flow with a global non-relativistic Coulomb unitary evolution. Section 5a proves the fixed-interval Kepler localization and both first-moment limits under its stated regularization and coherent-preparation hypotheses. Proposition 2 assigns endpoints to explicit selection criteria; it does not derive those criteria from Newton's axioms.

Still open: the planar operator beyond the cited form construction; critical full Klein–Gordon dynamics under the no-log prescription; long-time or shrinking-regularization estimates; and the necessity claim that every completion of Newton's mechanics requires positive h. Classical regularizations remain possible added prescriptions, and Dirac self-adjoint extensions exist at every $h > 0$ if boundary data are allowed. Excluding those routes requires an independently

justified premise. The quartic Newtonian initial-value problem is globally well posed; its contrast concerns the particular classical-first time-slicing construction. These limitations are part of the statement, not consequences to be inferred from a threshold formula.

7. Consequence for STATE

The fixed-T, fixed-δ localization obligation (L) and the Kepler centroid obligation are closed by the written derivation in §5a. Replace the former unqualified action interval by Proposition 2's selection-dependent sets: Dirac finite-potential selection $(k/c, \infty)$, Dirac critical-form/cutoff selection $[k/c, \infty)$, and Klein–Gordon strict-energy construction $(2k/c, \infty)$; its radial principal Friedrichs endpoint is included, with full critical dynamics still open. Next: decide which selection premises can be required independently of the desired action gap, then address the exclusion of completions without h. The October 1 referee acceptance applies to the original Theorem 1 as then printed. The October 2 localization, moment proof and replacement Proposition 2 are written, source-supported additions by Codex and have not received independent verification or referee acceptance.